

Hyperuniform isotope engineering of phonon transport: exact scaling laws and a stealthy transparency window in hexagonal boron nitride

Ronghua Wu¹, Ruqing Chen²

¹*Zhuhai Jiuchongtian Aviation Technology Co., Ltd., Zhuhai, China*

²*GUT Geoservice Inc., Montreal, Canada*

wrh@gdtzn.com, ruqing@hotmail.com

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Abstract

Isotopic substitution is one of the few strictly mass-like perturbations available in a crystal: within the Born–Oppenheimer approximation the interatomic force constants are invariant under isotope exchange, so an isotope pattern modifies the dynamical matrix through the mass factors alone. This makes the *spatial arrangement* of isotopes a design variable independent of the isotopic variance g_2 that has been the sole focus of isotope engineering to date. We generalize the Tamura theory of phonon–isotope scattering to spatially correlated arrangements by introducing the mass-fluctuation structure factor $S_\epsilon(\mathbf{k}) = g_2 s(\mathbf{k})$, and show that when $s(k) \simeq (k/k_0)^\alpha$ at small wavenumber the relaxation rate obeys the exact scaling law $\tau_{\text{iso}}^{-1} \propto \omega^{d+1+\alpha}$ in d dimensions, with closed-form angular prefactor $2/(\alpha + 2)$. The hyperuniformity exponent therefore contributes an additive, dimension-independent correction to the Rayleigh exponent, orthogonal to the dimensional baseline. We realize the class II case $\alpha = 1$ using sequences with Gaussian-unitary-ensemble spacing statistics—of which the unfolded nontrivial zeros of the Riemann zeta function are the canonical example—mapped onto a one-dimensional $^{10}\text{B}/^{11}\text{B}$ modulation of hexagonal boron nitride, obtaining $\tau_{\text{iso}}^{-1} \propto \omega^5$ in place of the Rayleigh ω^4 . We further find numerically that sequences built from zeros at finite height are more rigid than GUE, giving $\alpha = 1.49 \pm 0.12$, and that this exponent is independent of the height of the zeros used: across eight disjoint windows spanning $\bar{\gamma} = 1.2 \times 10^3$ to 6.2×10^7 the fitted trend is -0.138 ± 0.075 , consistent with zero, so the deeper suppression is available without any tuning. For stealthy (class I) arrangements we prove that the golden-rule rate vanishes identically, not merely asymptotically, for all $\omega < vK/2$, where K is the stealthiness wavenumber. We propose a differential transmission observable $\Delta\mathcal{T}(\omega)$ between concentration-matched samples that cancels the Casimir and anharmonic backgrounds to first order, identify a feasible observation window of 4–7 THz at liquid-helium temperature, and state the resulting fabrication specification—1.25 nm modulation pitch over 300 μm —as an explicit design target that exceeds present growth capability.

I. INTRODUCTION

Isotope engineering of phonon transport has, since the earliest work on diamond and silicon, been organized around a single scalar, the mass variance

$$g_2 = \sum_i f_i \left(1 - \frac{m_i}{\bar{m}}\right)^2, \quad (1)$$

with f_i the abundance of isotope i . Reducing g_2 by isotopic purification has produced record thermal conductivities in diamond, silicon, boron arsenide, and most strikingly in cubic boron nitride, where enrichment to 99% in either boron isotope yields $\kappa_{\text{RT}} > 1600 \text{ W m}^{-1}\text{K}^{-1}$ [1]. In hexagonal boron nitride the same strategy raises the in-plane conductivity to $585 \text{ W m}^{-1}\text{K}^{-1}$ [2] and lengthens hyperbolic phonon-polariton lifetimes substantially [3].

All of this treats the isotope distribution as *random*. The Tamura and Klemens theories that underpin it assume explicitly that isotopic occupations on distinct sites are uncorrelated [4, 5]; under that assumption the mass-fluctuation structure factor is white and the whole problem collapses onto g_2 .

There is no physical reason for this assumption to hold in a *grown* crystal. Layer-by-layer precursor switching permits, in principle, an arbitrary prescribed sequence. This paper asks what becomes available once the arrangement is treated as a design variable at fixed g_2 .

The natural language is hyperuniformity. A point pattern is hyperuniform when its structure factor vanishes in the infinite-wavelength limit, so that long-wavelength density fluctuations are anomalously suppressed [6, 7]. Torquato’s classification organizes hyperuniform systems by the exponent α in $S(k) \sim |k|^\alpha$: class I ($\alpha > 1$, all crystals and quasicrystals), class II ($\alpha = 1$, logarithmically growing number variance [8]), and class III ($0 < \alpha < 1$); the connection between random-matrix spectra, number theory and hyperuniform point processes is developed in Ref. [9]. Random arrangements correspond to $\alpha = 0$.

Long-wavelength phonons are sensitive to precisely the fluctuations that hyperuniformity suppresses. We establish below that the isotope-scattering exponent is controlled by α in a simple additive way.

To realize a specific α we require a sequence whose small- k structure factor is known exactly. The nontrivial zeros of the Riemann zeta function provide the canonical example. Under the Montgomery–Odlyzko correspondence [10, 11] the unfolded zeros have pair statistics identical to the eigenvalues of the Gaussian unitary ensemble (GUE), for which Dyson’s exact solution yields the closed form

$$s_{\text{GUE}}(k) = \begin{cases} k/2\pi, & k < 2\pi, \\ 1, & k \geq 2\pi, \end{cases} \quad (2)$$

in units of unit mean spacing. This is $\alpha = 1$, though only asymptotically in the height of the zeros; Sec. 3.3 quantifies the finite-height departure, which is substantial at any realizable length.

We emphasize that the zeros enter purely as a mathematically exact generator of a class II sequence. Every prediction below depends on the arrangement only through α , and any $\beta = 2$ level-repelling point process would serve identically. The zeros are chosen because their structure factor is known in closed form, not because of any presumed significance of the sequence itself.

II. THEORETICAL FRAMEWORK

A. Exactness of the mass perturbation

Within the Born–Oppenheimer approximation nuclear masses do not enter the electronic Hamiltonian. The second-order force-constant tensor

$$\Phi_{ij}^{\alpha\beta} = \frac{\partial^2 E}{\partial u_i^\alpha \partial u_j^\beta} \quad (3)$$

is therefore invariant, element by element, under isotopic substitution; only the mass factors in the dynamical matrix change,

$$D(\mathbf{q}) = M^{-1/2} \Phi(\mathbf{q}) M^{-1/2}. \quad (4)$$

Two consequences follow. Physically, an isotope pattern probes the phonon eigenvectors without deforming the potential-energy surface. Computationally, Φ need be evaluated only once, on the primitive cell; arbitrary arrangements follow at zero additional first-principles cost by permuting the diagonal matrix M (Sec. 3).

Writing $M_n = \bar{M}(1 + \epsilon_n)$ with $\epsilon_n \equiv \Delta M_n / \bar{M}$,

$$\delta D = -\omega^2 \sum_n \epsilon_n |n\rangle \langle n| + \mathcal{O}(\epsilon^2). \quad (5)$$

For the $^{10}\text{B}/^{11}\text{B}$ pair $|\epsilon| \lesssim 0.05$, so second-order contributions are suppressed by $\sim 10^{-3}$ relative; their role is bounded in Sec. 2.5.

B. Generalized Tamura formula

The matrix element between Bloch states is

$$\langle \mathbf{q}' | \delta D | \mathbf{q} \rangle = -\frac{\omega^2}{N} \sum_n \epsilon_n e^{i(\mathbf{q}-\mathbf{q}') \cdot \mathbf{R}_n}. \quad (6)$$

We define the *mass-fluctuation structure factor*

$$S_\epsilon(\mathbf{k}) \equiv \frac{1}{N} \left| \sum_n \epsilon_n e^{i\mathbf{k} \cdot \mathbf{R}_n} \right|^2 = g_2 s(\mathbf{k}), \quad (7)$$

with $s \rightarrow 1$ at large k . Fermi’s golden rule then gives

$$\frac{1}{\tau(\omega)} = \frac{\pi}{2} \omega^2 D(\omega) \langle S_\epsilon(|\mathbf{q} - \mathbf{q}'|) \rangle_{\text{FS}}, \quad (8)$$

where $\langle \cdot \rangle_{\text{FS}}$ denotes the average over the constant-frequency surface.

Equation (8) is the central relation of this work. Its only generalization relative to the standard Tamura result is the factor $s(\mathbf{k})$; setting $s \equiv 1$ recovers the uncorrelated theory exactly. Like the original it is a first Born treatment on a virtual-crystal background, whose limitations for strong or optical-branch scattering are analyzed in Ref. [12]; for the weak $^{10}\text{B}/^{11}\text{B}$ mass contrast considered here those corrections are negligible. The entire content of arrangement engineering resides in the small- k behavior of s .

Two remarks on validity. First, Eq. (8) is first order in ϵ and describes single-scattering processes; multiple scattering is addressed in Sec. 2.5. Second, in the multi-sublattice case the right-hand side acquires the eigenvector projection $\sum_\kappa g_2(\kappa) |e(\kappa|\mathbf{q}, j)|^2$ familiar from Tamura’s original treatment. In h-BN this factor is strongly asymmetric: with natural nitrogen (99.64% ^{14}N) the nitrogen sublattice contributes $g_2 \approx 1.8 \times 10^{-5}$, two orders of magnitude below an equimolar boron sublattice, so the boron projection dominates. We retain the scalar form and restore the projection where numerical values are quoted.

C. Classification of arrangements

Let

$$s(k) \simeq (k/k_0)^\alpha, \quad k < k_0, \quad (9)$$

with k_0 the crossover wavenumber above which s saturates. Random alloys give $\alpha = 0$; class III, $0 < \alpha < 1$; class II (GUE, Riemann zeros) $\alpha = 1$ with number variance $\sigma^2(R) \sim R^{d-1} \ln R$; class I (periodic, quasiperiodic, stealthy) $\alpha > 1$. For the zeta-zero mapping, Eq. (2) with physical spacing ℓ gives $k_0 = 2\pi/\ell$ and $\alpha = 1$ exactly.

D. Exact scaling law

Insert Eq. (9) into Eq. (8). In three dimensions the momentum transfer on the elastic shell is $k = 2q \sin(\theta/2)$, and with the substitution $u = \sin(\theta/2)$ the angular average evaluates in closed form,

$$\langle k^\alpha \rangle = (2q)^\alpha \int_0^\pi \sin^\alpha \frac{\theta}{2} \cdot \frac{1}{2} \sin \theta d\theta = (2q)^\alpha \frac{2}{\alpha + 2}. \quad (10)$$

Combining with the Debye density of states $D(\omega) \propto \omega^2$,

$$\frac{1}{\tau_{3D}(\omega)} = \frac{V_0 g_2}{4\pi v^3} \omega^4 \cdot \frac{2}{\alpha + 2} \left(\frac{2\omega}{vk_0} \right)^\alpha \quad (11)$$

valid for $\omega < vk_0/2$; above that frequency $s \rightarrow 1$ and the Rayleigh form is recovered. At $\alpha = 0$ the prefactor reduces to unity and Eq. (11) becomes the Klemens result exactly, which fixes the normalization without adjustable parameters.

Repeating the derivation in general dimension with $D(\omega) \propto \omega^{d-1}$ gives the compact statement

$$\boxed{\frac{1}{\tau_{\text{iso}}} \propto \omega^{d+1+\alpha}} \quad (12)$$

Hyperuniformity contributes an additive, dimension-independent correction $+\alpha$ to the Rayleigh exponent. Dimensionality sets the baseline (ω^4 in bulk, ω^3 in a monolayer, ω^2 in a quantized quasi-one-dimensional channel) and the arrangement supplies the correction; the two are orthogonal. Equation (12) is the principal analytical result of this work.

We note one caveat. Equation (12) assumes the Debye form of $D(\omega)$. In layered h-BN the acoustic dispersion is three-dimensional only below the interlayer crossover; above it the density of states approaches the two-dimensional form and the baseline exponent drifts from 4 toward 3. The additive $+\alpha$ is unaffected, since it originates entirely in $s(k)$, but the absolute exponent must be extracted from the computed $D(\omega)$ rather than assumed.

Setting $\alpha = 1$ and $k_0 = 2\pi/\ell$,

$$\frac{1}{\tau_{\text{II}}(\omega)} = \frac{V_0 g_2 \ell}{6\pi^2 v^4} \omega^5, \quad \mathcal{R}(\omega) \equiv \frac{\tau_{\text{rand}}}{\tau_{\text{II}}} = \frac{2}{3} \frac{\omega \ell}{\pi v}. \quad (13)$$

Three features deserve emphasis. (i) The suppression vanishes linearly as $\omega \rightarrow 0$ and possesses no threshold—the qualitative distinction from class I. (ii) $\mathcal{R} \propto \ell$: denser modulation gives *stronger* suppression, because $k_0 = 2\pi/\ell$ pushes the suppressed region to higher wavenumber. (iii) The crossover frequency $f_0 = v/2\ell$ is fixed entirely by the geometric pitch and is independent of the sequence; the number-theoretic input supplies only the exponent 1 and the prefactor $2/3$.

E. Stealthy arrangements: an exact theorem

Theorem. If the mass arrangement satisfies $S_\epsilon(\mathbf{k}) = 0$ for all $0 < |\mathbf{k}| < K$, then the golden-rule rate (8) vanishes identically for every $\omega < vK/2$.

Proof. On the elastic shell the momentum transfer obeys $|\mathbf{q} - \mathbf{q}'| \in [0, 2q]$ with $q = \omega/v$. If $\omega < vK/2$ then $2q < K$, so every accessible transfer lies in $(0, K)$, where S_ϵ vanishes by construction. The $\mathbf{k} = 0$ term is forward scattering: it does not relax momentum and contributes only a mean-mass renormalization absorbable into $\bar{M}$ and hence into v . The integrand of Eq. (8) therefore vanishes over the entire constant-frequency surface. ■

This is a genuine gap, not an asymptotic suppression:

$$\boxed{\omega_{\text{window}} < \frac{vK}{2}, \quad K = \chi k_0 = \frac{2\pi\chi}{\ell}} \quad (14)$$

with χ the stealthiness parameter. For a binary mass constraint in one dimension, collective-coordinate optimization [13] reaches $\chi \lesssim 0.5$, giving $\omega_{\text{window}} < \pi v/2\ell$.

Three limitations must be stated explicitly. (i) *First order only.* Second-order processes $\mathbf{q} \rightarrow \mathbf{q}'' \rightarrow \mathbf{q}'$ pass through off-shell intermediate states whose individual momentum transfers may exceed K while summing to a small net transfer; the residual rate is of relative order $g_2^2 \sim 10^{-6}$ —negligible, but not zero. (ii) *Finite-size floor.* A sequence of N sites can enforce $S_\epsilon = 0$ at at most $\sim \chi N$ wavevectors; between constrained points the residual is $\mathcal{O}(1/N)$. (iii) *Isotope channel only.* Stealthiness has no effect on anharmonic, point-defect, or boundary scattering. It suppresses one term of Matthiessen’s rule and leaves the others untouched—the central practical constraint of Sec. 4.

All three arrangement classes share the *same* g_2 —the same isotopic composition, differing only by permutation:

$$\frac{1}{\tau_{\text{iso}}} \propto g_2 \omega^4 \times \begin{cases} 1, & \text{random alloy,} \\ \frac{2}{3} \omega \ell / \pi v, & \text{class II,} \\ 0, & \text{class I, } \omega < vK/2. \end{cases} \quad (15)$$

This shared- g_2 property is what makes the differential observable of Sec. 4 possible.

III. COMPUTATIONAL METHODS

The pipeline requires one primitive-cell first-principles calculation and no large-matrix diagonalization at any stage.

A. Force constants

Second-order interatomic force constants are obtained for the two-atom h-BN primitive cell by density-functional perturbation theory, or equivalently by finite displacements, and transformed to real space, with the acoustic sum rule imposed after Fourier interpolation. By Eq. (3), Φ is independent of isotopic composition; all subsequent arrangements are constructed by rebuilding the diagonal mass matrix in Eq. (4), at $\mathcal{O}(N)$ cost and memory. The exponential supercell growth that would defeat a direct first-principles treatment of an aperiodic sequence does not arise.

B. Sequence generation and lattice mapping

Nontrivial zeros γ_n are located by the Riemann–Siegel formula with Gram-point bracketing [14], unfolded to unit mean spacing by the Riemann–von Mangoldt density,

$$\tilde{\gamma}_n = \frac{\gamma_n}{2\pi} \log \frac{\gamma_n}{2\pi e}, \quad (16)$$

computed by the Riemann–Siegel formula with Gram-point bracketing [14], and mapped onto the discrete sublattice by

$$z_n = a \left\lfloor \frac{\ell \tilde{\gamma}_n}{a} \right\rfloor, \quad (17)$$

where a is the lattice constant along the modulation direction, ℓ the target pitch, and $\lfloor \cdot \rfloor$ denotes rounding. Control sequences (random, periodic, Fibonacci, stealthy) are generated on the same lattice at identical composition. For numerical

work the class II sequence may equivalently be drawn from a GUE spectrum sampled by the Dumitriu–Edelman tridiagonal model [15], which is statistically identical at the level of pair correlations and requires no external data.

C. Discretization and the preservation of $\alpha = 1$

The rounding in Eq. (17) is the principal methodological concern: it superposes a random displacement on each site and could in principle restore a white background. Modeling the rounding error as an independent displacement u_n uniform on $[-a/2, a/2]$, the structure factor decomposes exactly as

$$s_{\text{disc}}(k) = |f(k)|^2 s_{\text{ideal}}(k) + [1 - |f(k)|^2], \quad f(k) = \text{sinc}\left(\frac{ka}{2}\right). \quad (18)$$

The second term is a non-hyperuniform background, but it is not white: expanding for $ka \ll 1$,

$$1 - \text{sinc}^2(ka/2) \simeq \frac{(ka)^2}{12}, \quad (19)$$

which vanishes as k^2 , i.e. *faster* than the ideal class II term k/k_0 . Discretization therefore does not destroy hyperuniformity; it adds a subleading contamination of relative size $\pi ka^2/6\ell$, which at the band edge $k = k_0 = 2\pi/\ell$ reduces to the parameter-free design rule

$$\left. \frac{s_{\text{bg}}}{s_{\text{ideal}}} \right|_{k_0} = \frac{\pi^2}{3} \left(\frac{a}{\ell} \right)^2 \quad (20)$$

falling linearly toward $k \rightarrow 0$. Requiring $\leq 15\%$ contamination at the band edge gives $\ell \gtrsim 5a$; with the h-BN in-plane lattice constant $a = 2.504 \text{ \AA}$ this is $\ell \gtrsim 1.25 \text{ nm}$ —a bound that coincides with the pitch selected on independent transport grounds in Sec. 4.3. We regard this coincidence as the natural operating point of the design.

Numerically, $s(k)$ is evaluated for every Fourier mode by FFT and band-averaged in logarithmically spaced bins. This step is essential: a raw periodogram ordinate is exponentially distributed, so a single realization on a sparse k grid yields fitted exponents with uncertainties of order ± 0.5 and an apparent α biased toward 0.65–0.89 by spectral leakage from the large- k plateau. With band averaging, a GUE spectrum sampled by the Dumitriu–Edelman model recovers $\alpha = 1.04 \pm 0.06$ and the random control $\alpha = 0.02 \pm 0.05$ at $N = 6 \times 10^4$ sites, while the number variance of the same unfolded spectra agrees with the GUE prediction $\sigma^2(R) = [\ln(2\pi R) + \gamma_E + 1]/\pi^2$ to better than one percent. We report both tapered and untapered exponents so that the size of the estimator artifact remains visible.

Sequences built from the actual zeta zeros behave differently, and the distinction matters. The Montgomery–Odlyzko correspondence is asymptotic in height and is approached only logarithmically; finite-height departures from GUE are well documented in the random-matrix and quantum-chaos literature [11, 16] as semiclassical prime-pair corrections to the spectral form factor. Odlyzko’s clean numerical agreement required zeros near 10^{20} , far beyond any realizable modulation length.

Figure 1 quantifies the consequence. We measure eight *disjoint* height windows – each averaging 24 blocks of 5352 zeros, with starting indices separated by more than the 1.28×10^5 zeros a single point consumes – spanning sampled heights $\bar{\gamma} = 1.2 \times 10^3$ to 6.2×10^7 .

Two statistics behave differently, and the contrast is the substance of the figure. The nearest-neighbour spacing dispersion σ_s rises monotonically in all eight windows, from 0.4028 to 0.4161 at $+0.0029$ per decade, and linear extrapolation reaches the GUE value 0.4180 only near $\bar{\gamma} \sim 10^{8.4}$: low-lying zeros are unambiguously more rigid than GUE, and the approach is slow. The exponent α , by contrast, does *not* track height. Its weighted mean over all eight windows is $\alpha = 1.49 \pm 0.12$, exceeding unity by 4.1σ , but a weighted fit against $\log_{10} \bar{\gamma}$ gives a slope of only -0.138 ± 0.075 (1.8σ ; likelihood-ratio $\Delta\chi^2$ agrees), and a constant α describes the data comfortably ($\chi^2/\text{dof} = 0.73$).

The apparent decline is carried entirely by the lowest window. Excluding it, the remaining seven points spanning $\bar{\gamma} = 1.4 \times 10^5$ to 6.2×10^7 are flat to within their errors: slope $+0.013 \pm 0.149$ (0.1σ), $\chi^2/\text{dof} = 0.07$ about their common mean $\alpha = 1.364 \pm 0.135$. The lowest window sits 2.2σ above that mean. Extending the lever arm from five to eight decades relative to a preliminary run *reduced* the fitted slope, from -0.234 to -0.138 , while leaving its significance unchanged – the signature of a spurious trend rather than an unresolved one.

We therefore report a definite negative result: above $\bar{\gamma} \sim 10^5$ the exponent of a lattice-mapped zeta sequence is constant at $\alpha = 1.36 \pm 0.14$, and height is *not* a usable design parameter for α . That σ_s drifts while α does not is itself informative: the short-range spacing statistic and the $k \rightarrow 0$ asymptotics that set α converge toward GUE at different rates, and only the former retains height sensitivity in the accessible range.

The physical implication is nonetheless favourable: $\alpha \simeq 1.4$ deepens the suppression of Eq. (12) from ω^5 to roughly $\omega^{5.4}$. Where a clean $\alpha = 1$ reference is required we use the GUE surrogate, which delivers $\alpha = 0.973 \pm 0.029$ under identical analysis.

A second systematic deserves note. Enforcing a target composition by flipping a fraction f of sites at random superposes a white component of relative size $\sim f/[c(1-c)]$ on $s(k)$; on a class I sequence $f \sim 10^{-3}$ already suffices to measure $\alpha \approx 0$. Every generator must therefore reach the target composition by construction.

D. Transport observables

For each transverse wavevector $k_{\parallel}^{(m)}$ an effective one-dimensional chain is constructed, retaining all interlayer and further-neighbor force constants without truncation. The implementation of every step below, including the Riemann–Siegel zero finder used for Eq. (16) and the window-disjointness constraint applied in Fig. 1, is available in the accompanying repository.

Three mutually redundant routes are used: (i) the smallest Lyapunov exponent $\gamma(\omega)$ from transfer-matrix products with

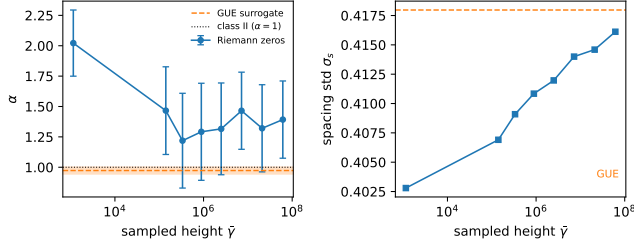


Figure 1: Finite-height statistics of lattice-mapped Riemann-zero sequences, from eight disjoint windows of 24×5352 zeros each. Left: hyperuniformity exponent α versus sampled height $\tilde{\gamma}$, with the GUE surrogate (dashed, shaded band) and the class II value $\alpha = 1$ (dotted). The weighted mean is $\alpha = 1.49 \pm 0.12$, but α shows no height dependence: the fitted slope is -0.138 ± 0.075 (1.8σ), it is carried entirely by the lowest window, and the upper seven points are flat ($+0.013 \pm 0.149$, $\chi^2/\text{dof} = 0.07$ about $\alpha = 1.364 \pm 0.135$). Right: nearest-neighbour spacing dispersion σ_s over the same windows, rising monotonically in all eight and extrapolating to the GUE value near $\tilde{\gamma} \sim 10^{8.4}$. The short-range rigidity drifts with height; the $k \rightarrow 0$ exponent does not.

periodic reorthogonalization, cost $\mathcal{O}(N)$ and the cleanest route to the exponent of Eq. (12); (ii) recursive Green’s function evaluation of the Landauer transmission $\mathcal{T}(\omega) = \sum_m \mathcal{T}_m(\omega)$; and (iii) the kernel polynomial method for $D(\omega)$, which also verifies that the class II sequence produces no spectral gaps—a negative result reported explicitly, since gap formation requires Bragg peaks that a GUE-class sequence does not possess.

IV. PROPOSED OBSERVABLE AND DESIGN SPECIFICATIONS

A. The differential measurement

At low temperature the total relaxation rate decomposes by Matthiessen’s rule,

$$\frac{1}{\tau_{\text{tot}}} = \underbrace{\frac{v}{L_{\text{eff}}}}_{\text{Casimir}} + \frac{1}{\tau_{\text{anh}}(\omega, T)} + \frac{1}{\tau_{\text{def}}(\omega)} + \frac{1}{\tau_{\text{iso}}(\omega; s)}. \quad (21)$$

Only the last term depends on the isotope arrangement. The Casimir boundary term [17, 18] is frequency independent and exceeds the isotope term throughout the sub-terahertz range, so any absolute measurement is background dominated.

Consider two samples carrying a class II and a random arrangement, with identical isotopic composition, geometry, surface preparation, growth batch, and point-defect density. Their Hamiltonians then differ by a permutation of masses alone; the first three terms of Eq. (21) are identical term by term, and

$$\Delta\left(\frac{1}{\tau}\right) = \frac{1}{\tau_{\text{iso}}^{\text{rand}}(\omega)} [1 - \mathcal{R}(\omega)]. \quad (22)$$

The composition-matching requirement is load bearing. A mismatch δc produces a spurious difference in g_2 ; near $c =$

Table 1: Feasibility of the differential observable at $T \lesssim 4$ K for $\ell = 1.25$ nm and $L = 300$ μm .

f (THz)	τ_{iso}^{-1} (s^{-1})	τ_B/τ_{iso}	$\mathcal{R}$	$\Delta\tau_{\text{tot}}^{-1}/\tau_{\text{tot}}^{-1}$
4	3.6×10^7	0.53	0.50	17%
5	8.8×10^7	1.3	0.63	21%
6	1.8×10^8	2.7	0.75	18%
7	3.4×10^8	5.0	0.88	10%
8	5.8×10^8	8.6	1.00	0

0.5 the leading order cancels but the second-order term still reaches the target effect for $\delta c \sim 1\%$. Compositions must be matched to $\delta c < 0.2\%$, verified by secondary-ion mass spectrometry and independently by the Raman E_{2g} position, which shifts from 1394 cm^{-1} in h^{-10}BN to 1358 cm^{-1} in h^{-11}BN [19].

Transmission is not linear in the relaxation rate. With $\mathcal{T}_m = (1 + L/\Lambda_m)^{-1}$ and $\Lambda_m = v_m \tau_m$,

$$\Delta\mathcal{T}(\omega) \simeq \sum_m \mathcal{T}_m^2 \frac{L}{v_m} \frac{1}{\tau_{\text{iso},m}^{\text{rand}}} [1 - \mathcal{R}(\omega)]. \quad (23)$$

The background cancels from the signal but not from the sensitivity: the prefactor $\mathcal{T}_m^2$ retains all background scattering, so the sample must remain quasi-ballistic, $L \lesssim \Lambda_{\text{background}}$.

B. Transverse integration

A natural objection is that integration over $k_{\parallel}$ washes out one-dimensional sequence effects. For Cantor-set gaps in quasiperiodic superlattices this is valid: gap positions migrate channel by channel and the summed spectrum is smeared. For power-law scaling it is not. The angular average of Eq. (10) is the continuum-limit channel sum; it preserves the exponent and modifies only the prefactor $2/(\alpha+2)$. Exponents are scale invariants and are immune to channel averaging.

This is why we adopt a thick three-dimensional geometry rather than a nanoribbon. Quantizing $k_{\parallel}$ is unnecessary here, and the edges required to achieve it introduce diffuse boundary scattering scaling as $\Delta^2 \omega^2 / vW$ (Appendix), which necessarily dominates $\omega^{4+\alpha}$ at low frequency.

C. Feasibility window

Taking in-plane h-BN parameters $v \simeq 2 \times 10^4 \text{ m s}^{-1}$ and $V_0 = 9.05 \text{ \AA}^3/\text{atom}$. An equimolar $^{10}\text{B}/^{11}\text{B}$ boron sublattice has $g_2^{\text{B}} = 2.245 \times 10^{-3}$, but only the boron projection of the phonon eigenvector contributes: for long-wavelength acoustic modes $|e(\kappa)|^2 = M_{\kappa}/\sum_{\kappa'} M_{\kappa'}$, giving a boron weight $10.81/(10.81 + 14.01) = 0.436$ and hence an effective $g_2 = 0.436 \times 2.245 \times 10^{-3} = 9.8 \times 10^{-4} \simeq 1 \times 10^{-3}$. With this, $\tau_{\text{iso}}^{-1} \simeq 9.0 \times 10^{-47} \omega^4 \text{ s}^{-1}$. Below ~ 4 K, Umklapp processes are frozen out and normal processes do not relax momentum, so the background is Casimir limited. For $\ell = 1.25$ nm ($f_0 = v/2\ell = 8$ THz) and $L = 300$ μm :

The accessible window is 4–7 THz, optimal near 5–6 THz, with a predicted differential signal $\Delta\mathcal{T}/\mathcal{T} \sim 20\%$. Both edges squeeze each other: the lower edge is set by the Casimir background, whose reduction requires larger L but is capped by the

quasi-ballistic condition; the upper edge is $f_0 = v/2\ell$, whose increase requires smaller ℓ but is bounded below by Eq. (20) and by the resolvability of the GUE spacing distribution on the discrete lattice. The two constraints intersect at $\ell \simeq 1.25$ nm, which we adopt as the design specification.

These values assume in-plane propagation and modulation. Modulation along the stacking axis is more accessible to layer-by-layer growth but carries $v_c \simeq 3.5 \times 10^3$ m s⁻¹, which shifts f_0 down by a factor ~ 6 and closes the overlap with the Casimir crossover.

D. Fabrication specification

The window requires isotopic placement at 1.25 nm pitch over 300 μ m: approximately 2.4×10^5 independently specified modulation units. We state plainly that no present h-BN growth technique meets this specification. Precursor switching in MOCVD along the c axis provides layer-level control but coherent modulation lengths of order 1 μ m; in-plane isotopic patterning at nanometer resolution has no established route.

This paper is therefore a theory and design-specification study. Its contributions are the scaling law of Eq. (12), the stealthy window theorem, the discretization bound of Eq. (20), and the quantitative delimitation of the observation window together with the growth capability required to reach it. We regard explicit statement of that gap as more useful than its concealment.

V. CONCLUSION

The spatial arrangement of isotopes, at fixed isotopic variance, is a genuine and previously unused control parameter for phonon transport. Generalizing the Tamura theory through the mass-fluctuation structure factor yields $\tau_{\text{iso}}^{-1} \propto \omega^{d+1+\alpha}$, in which the hyperuniformity exponent contributes additively and independently of dimension. Mapping a GUE-class sequence onto a ¹⁰B/¹¹B modulation converts the Rayleigh ω^4 into ω^5 ; sequences built from Riemann zeros at finite height are more rigid still, with $\alpha = 1.49 \pm 0.12$ independent of the height sampled, and a correspondingly deeper suppression. For stealthy arrangements the golden-rule rate vanishes identically below $vK/2$, defining a hard transparency window.

Hexagonal boron nitride is the natural host: its boron sublattice carries an anomalously large natural mass disorder ($g_2 \approx 1.35 \times 10^{-3}$, some eighteen times that of natural carbon) while its nitrogen sublattice is nearly isotopically pure, and its force constants are strictly untouched by the substitution.

Data availability. All code, generated sequences, and the manuscript source are openly available at <https://github.com/Ruqing1963/hbn-hyperuniform-isotope>. The repository reproduces every figure and every number quoted in Secs. 3 and 4 from a single primitive-cell force-constant calculation, and includes the regression tests for the scaling exponents of Eq. (12).

I. BOUNDARY ROUGHNESS SCALING

With the Ziman specular parameter $p(\omega) = \exp(-4q^2\Delta^2)$ for rms roughness Δ , the diffuse boundary rate is

$$\frac{1}{\tau_{\text{edge}}} = [1 - p(\omega)] \frac{v}{W} \xrightarrow{q\Delta \ll 1} \frac{4\Delta^2}{vW} \omega^2. \quad (24)$$

Comparison with Eq. (12) shows that boundary scattering dominates at low frequency whenever $d + 1 + \alpha > 2$, i.e. always. Equating the two rates gives the crossover width

$$W_{\text{cross}} = \frac{16\pi\Delta^2 v^2}{V_0 g_2 \omega^2}, \quad (25)$$

which for $\Delta = 0.5$ nm and $f = 1$ THz is of order 10^2 μ m. Any geometry narrow enough to quantize $k_{\parallel}$ is boundary dominated by four to six orders of magnitude, which excludes the nanoribbon route quantitatively.

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